



Oguz Umut SALMAN

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PROFESSIONAL EXPERIENCE



CNRS, Chargé de recherche

01.2015 – PRESENT

Researcher - Centre National Research Scientifique (CNRS) in Mechanics, FRANCE Lecturer - Université Sorbonne Paris Nord, FRANCE .



CNR-ieni , Scientific researcher

09.2013 – 09.2015

Post-Doctoral Fellow - CNR-ieni Milano, ITALY Advanced Grant of the European Research Council: Size Effects in Fracture and Plasticity .



BOSPHORUS UNIVERSITY, Post-doc

10.2011 – 10.2012

Post-Doctoral Fellow - The Physics Department under the supervision of M. Mungan.



HARVARD UNIVERSITY, Post-doc

10.2012 – 09.2013

Post-Doctoral Fellow - The Aisenberg Biomineralization and Biomimetic Laboratory, Harvard University, USA.



ECOLE POLYTECHNIQUE, Post-doc

06.2009 – 10.2011

Post-Doctoral Fellow - Solid Mechanics Laboratory (LMS), Ecole Polytechnique, FRANCE

EDUCATION



PHD. IN PHYSICS

09.2005 – 06.2009

Université Pierre et Marie Curie, Paris, FRANCE (High Honour Mention).



MSC. IN MATERIALS SCIENCE AND NANO-OBJECTS

09.2004 – 09.2005

Ecole Polytechnique, Palaiseau, FRANCE (High Honour Mention).



B.S. IN PHYSICS

09.2000 – 06.2004

Istanbul Technical University, Physics Department, Istanbul, TURKEY (Honour List).

SKILLS & INTERESTS



Native Speaker



Fluent



Fluent

PROGRAMMING: C, C++, Fortran, Mathematica, Matlab,
Parallel coding (OpenMP-MPI), ABAQUS ,
FreeFem++, GPU Programming

PUBLICATIONS

- O.U. Salman, M Ghidelli, J. Smiri and I.R. Ionescu, "*Computing the true stress in micro-pillars compression experiments* ", Submitted to Scripta Materials, (2023)
- R. Baggio, O.U. Salman, L. Truskinovsky, "*Nucleation of dislocations by pattern formation*", European Journal of Mechanics-A/Solids 99, 104897, (2023)
- R. Baggio, O.U. Salman, L. Truskinovsky, "*Inelastic rotations and pseudoturbulent plastic avalanches in crystals* ", Phys. Rev. E 107, 025004 – Published 24 February (2023)
- C. Baruffi, A. Finel, Y. Le Bouar, B. Bacroix, O.U. Salman, "*Atomistic simulations of temperature-driven microstructure formation in pure Titanium* ", Computational Materials Science 203, 111057, Editor's choice, (2022)
- O.U. Salman, I.R. Ionescu, "*Tempering the mechanical response of FCC micro-pillars: an Eulerian plasticity approach* ", Mechanics Research Communications, Volume 114, June , (2021)
- O.U. Salman, L. Truskinovsky, "*Delocalizing fracture* ", Journal of the Mechanics and Physics of Solids 154, 104517 , (2021)
- O.U. Salman, R Baggio, B Bacroix, G Zanzotto, N Gorbushin, "*Discontinuous yielding of pristine micro-crystals* ", Comptes Rendus. Physique 22 (S3), 1-48 , (2021)
- P. Zhang, O.U. Salman, J. Weiss, L. Truskinovsky, "*Fluctuations in crystalline plasticity* ", Comptes Rendus. Physique 22 (S3), 1-3707, (2021)
- P. Zhang, O.U. Salman, J. Weiss, L. Truskinovsky, "*Variety of scaling behaviors in nanocrystalline plasticity* ", Phys. Rev. E, 102, 023006, arXiv:2004.08579, hal-02644607, (2020)
- Edited by I.R. Ionescu, S. Queyreau, C. R. Picu, O.U. Salman, "*Mechanics and Physics of Solids at Micro- and Nano-Scales* ", ISBN: 978-1-786-30531-2 February (2020)
- O.U. Salman, R. Baggio, "*Homogeneous Dislocation Nucleation in Landau Theory of Crystal Plasticity*", appeared in the book *Mechanics and Physics of Solids at Micro- and Nano-Scales* , <https://doi.org/10.1002/9781119687566.ch1>, (2019)
- R. Baggio, E. Arbib, P. Biscari, S. Conti, L. Truskinovsky, G. Zanzotto, and O.U. Salman, "*Landau-Type Theory of Planar Crystal Plasticity* ", Phys. Rev. Lett. 123, 205501, (2019)
- O.U. Salman, G. Vitale, L. Truskinovsky, "*Continuum theory of bending-to-stretching transition*", Phys. Rev. E 100, 051001(R), Rapid Communication (2019)
- O.U. Salman, B. Muir, A. Finel, "*Origin of stabilisation of macro-twin boundaries in martensites*", The European Physical Journal B 92 (1), 20, (2019)
- C. Baruffi, A. Finel, Y. Le Bouar, B. Bacroix, O.U. Salman, "*Overdamped Langevin Dynamics simulations of grain boundary motion*", Materials Theory, 3,4 (2019)
- P. Franciosi, M. Spagnuolo, O.U. Salman, "*Mean Green operators of deformable fiber networks embedded in a compliant matrix and property estimates* ", Continuum Mechanics and Thermodynamics, Volume 31, Issue 1, pp 101–132, (2019)
- A. Ask, S. Forest, B. Appolaire, K. Ammar, O.U. Salman, "*A Cosserat crystal plasticity and phase field theory for grain boundary migration*", Journal of the Mechanics and Physics of Solids 115, 167-194, (2018)
- P. Zhang, O.U. Salman, J.-Y. Zhang, G. Liu, J. Weiss, L. Truskinovsky and J. Sun, "*Taming intermittent plasticity at small scales*" Acta Mater, 128, 351-364, (2018)

M. Jedrychowski, B. Bacroix, O.U. Salman, J. Tarasiuk, S. Wronski, "*Investigation of SIBM driven recrystallization in alpha Zirconium based on EBSD data and Monte Carlo modeling*", Materials Science and Engineering 89, 012029, (2016)

A. Taloni, O.U. Salman, S. Zapperi, "*Volume Changes During Active Shape Fluctuations in Cells*", Phys. Rev. Lett. 114, 208101, (2015) - (Highlighted in Nature Physics and APS Physics Journals)

O. Schylgo, O.U. Salman, A. Finel, "*Martensitic phase transformations in Ni-Ti base shape memory alloys: Landau Theory*", Acta Materialia 60, 6784-6792, (2012)

O.U. Salman, A. Finel, R. Delville, D. Schryvers, "*The role of phase compatibility in martensite*", J.Appl.Phys.111,103517, (2012)

O.U. Salman, L. Truskinovsky, "*On the critical nature of the plastic flow: one and two dimensional models*", International Journal of Engineering Science, 59:219.254, (2012)

O.U. Salman, L. Truskinovsky "*Minimal discrete automaton behind crystal plasticity*", Phys. Rev. Lett. 106, 175503, (2011)

A. Finel, O.U. Salman, "*Phase Field Methods: microstructures, mechanical properties and complexity*", Comptes Rendus Physique, Vol. 11, Issues 3-4, p. 245-256, (2010)

O.U. Salman, "*Criticality in martensite*", Mathematisches Forschungsinstitut Oberwolfach, DOI: 10.4171/OWR/2009/57 (2009)

R. Delville R. D. James, O.U. Salman, A. Finel and D. Schryvers "*Transmission electron microscopy study of low-hysteresis shape memory alloys*", DOI:10.1051/esomat/200902005, ESOMAT 2009, 02005 (2009)

B. Muite, O.U. Salman, "*Computations of geometrically linear and nonlinear Ginzburg-Landau models for martensitic pattern formation*", in ESOMAT 2009 edited by E. Sciences, P. Sittner, V. Paidar, L. Heller, H. Seiner, 587 p. 03008, (2009)

S. Ugur, O.U. Salman, G. Tepehan, F. Tepehan, O. Pekcan, "*Fluorescence study on Al_2O_3 polystyrene latex composite film formation*", Polymer composites 26 (3), 352-360, (2005)

CONFERENCES

International Conference on processing and manufacturing of advanced materials, Invited speaker, July 2 - 7, 2023 Vienna, Austria

"Plastic Turbulence", Invited speaker, Université Pierre et Marie Curie, Institut d'Alembert, 18 May (2022)

"Mécanique des Matériaux Architecturés", Invited speaker, Aussois, 17-21 January (2022)

"Landau theory of crystal plasticity: Application to nano-pillar plasticity", Invited speaker, Lund University, Mechanics Department, 15 October (2021)

"Avalanche dynamics and precursors of catastrophic events" 3-8th of February, Les Houches, France, (2019)

"11th US-France Mechanics and physics of solids), Speaker/Organiser, Paris, France, (2018)

"Phase Transformations and Plasticity in Crystals: Atomistic to Continuum Models", Invited Speaker, Politecnico di Milano (<https://www.fisi.polimi.it/it/home/news/57090>), Milano, September 3-4, (2018)

"Mesoscopic model of dislocational plasticity accounting for crystal symmetry", ECCM - ECFD 2018 6th European Conference on Computational Mechanics (Solids, Structures and Coupled Problems), Glasgow, UK, (2018)

"Mesoscopic model of dislocational plasticity accounting for crystal symmetry", The 13th World Congress in Computational Mechanics, NY, USA, (2018)

"The Landau theory of crystal plasticity ", Mechanics and physics of solids at micro- and nano-scales, 11th US-France Symposium, Paris, (2018)

HIGHLIGHTS

The paper *Landau-Type Theory of Planar Crystal Plasticity* published in the journal Physical Review Letters based on the Ph.D. work of R. Baggio that I co-supervised is highlighted in the journal cover. The same work is also highlighted on the website of INSIS:

<https://insis.cnrs.fr/fr/cnrsinfo/un-nouveau-modele-pour-etudier-la-plasticite-des-cristaux>



The paper *De-localisation of brittle fracture*, published in JMPS is highlighted on the website of INSIS:

<https://www.insis.cnrs.fr/fr/cnrsinfo/eviter-la-rupture-dun-materiau-en-redistribuant-la-deformation>

The paper *Cosserat crystal plasticity and phase field theory for grain boundary migration*, published in JMPS based on the post-doc work of A. Ask that I co-supervised is highlighted on the website of INSIS:

<http://www.cnrs.fr/insis/recherche/actualites/2018/09/joints-grains-metal.htm>

Received funding from Lundi University Mechanics Department for a stay of two months to work with A. Ahadi, (2021).

Received funding from Bonn University for a stay of two weeks to work with S. Conti and M. Mungan, (2019).

TEACHING

Visiting Professor – Lund University Mechanical Department, Sweden, (2022)

Invited Professor – Romanian-French Summer School in Applied Mathematics, Sinaia, ROMANIA, (2018)

Lecturer - Modelling of Functional Materials Master 2 Course (21H): Statistical mechanics, Theory of elasticity and Finite Element theory, ABAQUS Modelling Master 2 Course (21H), Université Sorbonne Paris Nord, FRANCE, (2015 - Present)

FUNDING

ANR: MESOCRYSP - *Discrete Mesoscale Plasticity*, 450000€ for LSPM, **Project Partner, Leader of LSPM team, collaboration PMMH-Centre des Matériaux**, (11/2021-09/2025)

ANR-JCJC: ALIS - *Modeling of reconstructive phase transformations through Atomistically informed Landau theory with Infinite Symmetry*, 160000€ for LSPM, **Project leader**, (03/2020-03/2023)

ANR-FWF: NANOFILM - *NANOARCHITECTED FILMS FOR UNBREAKABLE FLEXIBLE ELECTRONICS* (NANOFILM), 572K€ in total, 142K€ for LSPM, Collaboration with PPRIME-CNRS, ESI Leoben (Austria), I will lead the

phase-field modelling of fracture of thin films and supervise one post-doc, (03/2021-03/2024)

ANR-PRCI-SUMMIT - Collaboration with Xian Jiaotong University, (29 500€ for LSPM (265500€ total) Euro), Project partner, (01/2018-12/2021)

Labex-Projet émergent - Collaboration with M. Durant of Paris 7 University, UNified multi-scale framework for PLASticity of metals, amorphous and cellular materials, Project leader, (10/2018-10/2019)

ANR-I2T2M, Project partner, (3K€ total (01/2019-12/2022), **PICS-Pologne** - International Scientific Collaboration Project, CNRS (15K€), Project manager, (2017 – 2020), **Visiting Professor** - Funding Paris 13 University (2,5K Euro), (2017), **Concours coup de pouce**- F2M – French Mechanics Federation (50K €), Project Leader, (2015 - 2016)

SUPERVISIONS

Current

Post-Doc: **K. Ghosh**, Modelling of coupling of phase transition with crystal plasticity in Zirconium and Titanium through Landau theory , funded by ANR-JCJC, (18 Months), 2022-2024,

Post-Doc: **M. Terzi**, A low dimensional representation of 6D metric space via machine learning for the development of a new 3D Landau theory of crystal plasticity, funded by ANR-MESOCRYSP, (24 Months), 2021-2023,

Ph.D: **J. Smiri**, Eulerian finite element modelling of ductile fracture in micro-scale crystals, funded by Funded through Paris-Nord Sorbonne Université Scholarship, co-encadrant (36 Months), 2021-2024,

Ph.D: **M. Aissaoui**, Acoustic emission study of intermittent plasticity: the role of disorder, crystal orientation and symmetry, funded by Paris-Nord Sorbonne Université Scholarship, co-encadrant (36 Months), 2021-2024,

Ph.D: **G. Tejedor**, Modélisation numérique de la plasticité mésoscopique des cristaux par l'élasticité non linéaire invariante $GL(3, Z)$, funded by ANR-MESOCRYSP, co-encadrant (36 Months), 2022-2025

Completed

Post-Doc **M. Degeiter**, Modelling of Fracture in thin films through phase-field approach, funded by Labex, 2019-2022,

Post-Doc **N. Gorbushin**, Landau-type modeling of marginal stability in plasticity, funded by ANR-PRCI-SUMMIT, 2019-2021

Post-Doc **N. Perchikov**, Reduction to an automaton of geometrically nonlinear Landau theory of crystal plasticity, funded by ANR-PRCI-SUMMIT, 2019-2020

Ph.D, **R. Baggio**, Multi-Scale Modelling of Plasticity, Funded through Paris-Nord Sorbonne Université Scholarship, co-encadrant, defended in December, 2019, (post-doc in INRIA-Bordeaux and Université de Corse)

Ph.D, **M. Spagnuolo**, Modélisation continue des structures pantographiques, European- Inspire Scholarship, EU, co-encadrant, defended in September 2019

Post-Doc, **A. Ask**, Modélisation de Recristallisation Concours coup de pouce, Fédération francilienne de mécanique, FRANCE, funded by Concours coup de pouce de F2M, (2015-2016), (currently hired as research engineer in ONERA)

Ph.D, **C. Baruffi**, Modélisation des mécanismes de vieillissement et d'endommagement dans les alliages métalliques, ONERA Scholarship, FRANCE, defended in December 2018 (currently post-doc in EPFL Switzerland, **received le prix Jacques Dalla Torre 2019 for her Ph.D. work**)

Ph.D, **P. Zhang**, supervision of the last year of his Ph.D. work in Paris , collaboration China/France, defended in 2018, (currently hired as assistant professor in Xi'an Jiaotong University)

Stage M2, **F. Mansouri**, PNUS/LSPM (2016), **O. Benamor**, PNUS/LSPM (2020)

REVIEWER TASKS

I reviewed 15 articles between 2017-2022 submitted in journals: **Physical Review Letters, Physical Review B, Physical Review E, Physical Review X, Physical Review Materials, Computational Materials Science, Philosophical Magazine, Mechanics of Materials**.

I joined the Editorial Board of Interdisciplinary Physics (specialty section of Frontiers in Physics) as a Review Editor since 2021.

The C++ code implementing the crystal plasticity model based on $GL(2,Z)$ invariance is available:

<https://bitbucket.org/oguzumut-cnrs/cute-crystal/src>

The C++ code implementing the phase field model of fracture in thin films is available:

<https://bitbucket.org/oguzumut-cnrs/uspfn-fracture-fourier/src>

RESUMÉ EN FRANÇAIS

L'extrême miniaturisation de la technologie moderne exige une meilleure compréhension de la mécanique non conventionnelle, dominée par les fluctuations, des matériaux à des échelles micro et nanométriques. A ces échelles, la nature de la plasticité médiée par les dislocations change radicalement : les échantillons métalliques submicroniques présentent une limite d'élasticité élevée, cependant corrompue par des fluctuations de déformation intermittentes compromettant les processus de mise en forme et mettant en danger la stabilité structurelle. L'un des objectifs de ma recherche est de mieux comprendre ces phénomènes en développant des modèles mathématiques en allant au delà des outils de la mécanique continue classique comme la théorie de Landau en grande déformation qui peut représenter adéquatement les symétries particulières des réseaux cristallins ou le milieu de Cosserat. L'une de ma contribution majeure concerne la mise en évidence et la caractérisation de l'intermittence des événements résultant du comportement collectif des défauts (dislocations, joints de grains, macles). Il apporte ainsi une explication physique aux observations expérimentales les plus récentes réalisées sur les grands instruments (ESRF Grenoble). L'innovation de la contribution tient à la conjugaison d'approches discrètes et continues pour aborder la description des phénomènes critiques lors de la déformation de la matière. J'ai pour cela développé une mécanique des milieux continus raffinée, combinée à la méthode des champs de phase. Le caractère générique de ces avancées scientifiques se manifeste par les applications variées aux matériaux de l'ingénieur : métaux et alliages, matériaux composites (fibres longues, matrice polymère) matériaux amorphes et cellulaires.